Alice

Bob

S (TTP)

k_{AS}

k_{BS}

k_{AS}, k_{BS}

new N_A

A, N_A

new N_B

$B, N_B, \{A, N_A\}\ k_{BS}$

new k_{AB}

$N_B, \{B, k_{AB}, N_A\}\ k_{AS}, \{A, B, k_{AB}, N_B\}\ k_{BS}$

$\{A, B, k_{AB}, N_B\}\ k_{BS}, \{N_B\}\ k_{AB}$